

continuation

Continuation Document

Revision: 99 **Last modified:** 2026-06-02T01:13:11Z **Description:** Sacred invariant resumption document for Helix Cluster OS **Authority:** Constitution §12.10
Maintainer: Operator + AI loop

§0: How to use this document

Any CLI agent resuming work on this project MUST read this file first.

§1: Recently Completed Work (last 10 commits)

Commit	Message
0346e1c	Foundation wave 32: HXC-1334 HLC drift clamp implemented + 7 phase-6/5 items verified-already-done. HXC-1334 (pkg/hlc): added DefaultMaxDrift const + NewWithMaxDrift + a drift-clamp block in Clock.Update — a remote physical timestamp exceeding local wall-clock by >maxDrift is clamped to local+maxDrift (prevents far-future-remote clock poisoning, HLC's classic failure mode); Now() untouched, existing Update in-drift behaviour + 14 existing tests intact, 6 new tests; mutation neuter-clamp -> far-future remote (1.5e9) > ceiling (1.05e9) FAILS. VERIFIED already-complete (build+test+mutation-confirmed suites, NOT rebuilt); HXC-1335 pkg/crdt GCounter, HXC-1336 LWWRegister (timestamp+replica LWW tiebreak), HXC-1337 ORSet (mutation ignore-tombstones fails AddRemove tests), HXC-1338 pkg/crdt/merkle anti-entropy Diff (mutation Diff-nil fails), HXC-1340 pkg/federation Cell model+FSM (mutation

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caabf37	always-legal-transition fails RejectsIllegalJumps), HXC-1341 CellRegistry Register/Deregister/Lookup/ List/Advance thread-safe, HXC-1304 pkg/ inference LocalBackend wrapping internal/ llm. ZERO new deps. Gate: build/vet/vet- integration clean, full -short -race green, dataplane ok, tmux clean; drift-clamp + 3 batch mutations bite. Foundation wave 31: mark 3 items Completed (HXC-1290/1292/1297) — device Tier T1-T15 + AssignTier registry classifier / tier-aware+power-aware scheduler Plugin (verified). Mutation spot- checks confirm bites. Registry 205->208 Completed / 431 Queued. Foundation wave 31: HXC-1290+1292 combined + HXC-1297 verified-done. HXC-1290 (device universal Descriptor w/ 15 tiers) + HXC-1292 (device.AssignTier decision tree) — ONE deliverable: extended pkg/device.Tier enum additively to T1-T15 (+String cases; existing TierUnknown/T1- T8/ClassifyTier/EffectiveCompute byte-for- byte UNCHANGED, verified) + NEW pkg/ device/assign_tier.go AssignTier(d,reg *tierdef.Registry) Tier mapping Descriptor- >tierdef.DeviceCaps and iterating HIGH- >LOW returning the highest tier whose tierdef.Meets passes (networkUnknown=MaxInt sentinel so missing net field never disqualifies); 46 tests incl exact-per-tier T1/T8/T12/T15, exhaustive boundary sweep, monotonic property; mutation low->high iteration -> T15-spec wrongly returns T1 FAILS. Import device->tierdef (no cycle). HXC-1297 pkg/ scheduler tier-aware+power-aware Plugin VERIFIED already complete (tier_power.go min-tier filter + watts/perf-per-watt/ overshoot score + edgeaware.go + handheld.go, 67+ mutation-guarded tests; mutation tier-min < -> > fails TestTierPowerFilterRejectsBelowMinTier). ZERO new deps. Gate: build/vet/vet- integration ./... clean (tier.go edit no regression), full -short -race green, dataplane ok, tmux clean; iteration-order + tier-min mutations bite.
9baae11	Foundation wave 31: HXC-1290+1292 combined + HXC-1297 verified-done. HXC-1290 (device universal Descriptor w/ 15 tiers) + HXC-1292 (device.AssignTier decision tree) — ONE deliverable: extended pkg/device.Tier enum additively to T1-T15 (+String cases; existing TierUnknown/T1- T8/ClassifyTier/EffectiveCompute byte-for- byte UNCHANGED, verified) + NEW pkg/ device/assign_tier.go AssignTier(d,reg *tierdef.Registry) Tier mapping Descriptor- >tierdef.DeviceCaps and iterating HIGH- >LOW returning the highest tier whose tierdef.Meets passes (networkUnknown=MaxInt sentinel so missing net field never disqualifies); 46 tests incl exact-per-tier T1/T8/T12/T15, exhaustive boundary sweep, monotonic property; mutation low->high iteration -> T15-spec wrongly returns T1 FAILS. Import device->tierdef (no cycle). HXC-1297 pkg/ scheduler tier-aware+power-aware Plugin VERIFIED already complete (tier_power.go min-tier filter + watts/perf-per-watt/ overshoot score + edgeaware.go + handheld.go, 67+ mutation-guarded tests; mutation tier-min < -> > fails TestTierPowerFilterRejectsBelowMinTier). ZERO new deps. Gate: build/vet/vet- integration ./... clean (tier.go edit no regression), full -short -race green, dataplane ok, tmux clean; iteration-order + tier-min mutations bite.
9327930	Foundation wave 30: mark 3 items Completed (HXC-1302/1299/1291) — wireguard spot-drain teardown / discovery

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e4f0b03	15-tier registry / device.Probe real host detection. Mutation spot-checks confirm bites. Registry 202->205 Completed / 434 Queued. Foundation wave 30: 3 disjoint streams (all approved no-fix; mutation spot-checks bite). HXC-1302 wireguard preemption-safe teardown (pkg/wireguard/teardown.go: MeshCoordinator.TearDownPeers Stop+RemovePeer-all idempotent + SpotDrainHook(mc) cloudspot.Hooks.StopAdmission; tests w/ real curve25519 keys: add N -> teardown -> ListPeers==0, idempotent, FakeSource+Handler.Run drain removes peers; mutation skip-RemovePeer fails). HXC-1299 discovery 15-tier registry (pkg/discovery/tier_registry.go: TierRegistry.Register+SelectByTier(minTier) rank>=min via tierdef.Default() T1-T15 ordering, ordered exact-set; mutation >=<-> fails TestSelectByTier_T8 got T1/T5 want T8/T12/T15). HXC-1291 device.Probe (pkg/device/probe.go + probe_{darwin,linux,other}.go: real Arch(GOARCH)/CPUCores(NumCPU)/ MemoryBytes(sysctl hw.memsize / syscall.Sysinfo / MemStats fallback) + REAL os.Hostname-based ID, honest 0 for GPU/NPU/power; tests assert vs runtime oracle + real sysctl). ZERO new deps, ZERO edits to existing files. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok, tmux clean; B mutation bite + A/C reviewer-verified.
1e88aec	Foundation wave 29: mark 3 items Completed (HXC-1298/1296/1310) — handheld scheduling policy / device<->pb mapping (honest lossy) / per-tier security model. Mutation spot-checks confirm bites. Registry 199->202 Completed (crossed 200) / 437 Queued.
441c7fe	Foundation wave 29: 3 disjoint streams (all approved; A minor dead-helper + comment-lie fixed by me). HXC-1298 power/battery/thermal-aware handheld scheduling policy (pkg/scheduler/handheld.go HandheldPolicy Plugin reusing edgeaware battery/thermal/charging helpers: filter low-battery/over-thermal/unavailable/discharging-below-floor, score by battery/thermal/charging; absent-label assume-safe; 22 tests; mutation pct< ->

Commit	Message
634aa09	pct> fails LowBatteryRejected). HXC-1296 device.Descriptor<->pb.Node round-trip mapping (NEW pkg/devicemap: DescriptorToNode/NodeToDescriptor over MAPPABLE fields; HONEST lossy – round-trip identity over mapped subset + explicit test that device-only fields NPU/power/battery/trust are LOST (no false-recovery bluff); no import cycle; 7 tests). HXC-1310 per-tier security model enforcement (NEW pkg/tiersec reusing sandbox.Capability+edge+tierdef: ProfileForTier + Enforce(tier,op)->Decision over capability/data-class(inclusive)/network-egress CIDR allowlist via stdlib net; low-tier denied / high-tier allowed both directions; 24 tests; mutation data-class > -> >= fails inclusive-boundary). ZERO new deps, ZERO edits to existing files. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok, tmux clean; A+C mutations bite, B honest-loss reviewer-verified. Foundation wave 28: mark 3 items Completed (HXC-1311/1301/1303) — T1-T15 tier definitions / cloudspot PreemptionWatcher (verified) / inference Backend+Router (verified, dual-layer mutation). Registry 196->199 Completed / 440 Queued.
d4829d1	Foundation wave 28: HXC-1311 implemented + HXC-1301/1303 verified-already-done. HXC-1311 T1-T15 tier definitions (NEW pkg/tierdef: //go:embed tierdef.yaml with all 15 ascending tiers + min_requirements; LoadTiers parse+validate (rejects bad schema_version, missing/dup/out-of-order/wrong-count), Default()); TierDef.Meets(DeviceCaps) inclusive >= per-dimension (cpu/mem/gpu/vram/net); 73 test cases assert exact parsed T1/T15 values + per-dimension below-boundary false + exact-boundary true; mutation < -> <= fails TestMeets_ExactBoundary_True). HXC-1301 pkg/cloudspot PreemptionWatcher VERIFIED already complete (injectable PreemptionSource + 3-step graceful drain StopAdmission/Checkpoint/Deregister + deadline ForceStop + FakeSource + 9 mutation-killing tests). HXC-1303 pkg/inference provider-agnostic Backend interface + Router registry

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5745f9f	VERIFIED already complete (capability-aware Candidates/Route fallback + ReferenceBackend labeled output + ErrUnsupported; DUAL-LAYER capability correctness confirmed by me via mutation: neutering BOTH the Candidates filter AND the backend model gate fails TestRouterPicksCapabilityMatch/UnsupportedErrors – defense-in-depth, not a bluff). ZERO new deps. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok, tmux clean; 1311 boundary mutation + 1303 dual-layer mutation bite. Foundation wave 27: mark 2 items Completed (HXC-1237/1278) — swim prod+sim transport parity (real protocol over deterministic SimNetwork) / property-based fuzzing (4 Go fuzz targets). Mutation spot-checks confirm bites. Registry 194->196 Completed / 443 Queued.

§2: Environment Snapshot

Property	Value
Branch	main
Commit	0346e1c
Timestamp	2026-06-02T01:13:11Z

§3: Active Work

HXC	Title	Status
HXC-922	Node service, advisory locks, chaos tests, docs	✓ Done
HXC-921	Web UI, K8s/Helm, integration, E2E, benchmarks	✓ Done
HXC-920	Health, security, build, wireguard services	✓ Done
HXC-919	htmux, messaging, GPU, WASM, storage, build manifest	✓ Done
HXC-918	Wire scheduler/session gRPC to real backends, etcd discovery	✓ Done
HXC-916		✓ Done

HXC	Title	Status
	Stub expansion (config, retry, ratelimit, validator, build cache)	
HXC-914	LLM, policy, setup, distributed lock	✓ Done
HXC-913	Health, security, ClassAds parser	✓ Done
HXC-912	Gateway, helixd, helix-agent	✓ Done
HXC-911	Scheduler, Session gRPC services	✓ Done
HXC-910	Console, security, testing infra	✓ Done

§4: Next Planned Work

All MVP, Phase 2, Phase 3, and Phase 4 are **COMPLETE**. Potential future enhancements:

1. **Performance optimization** — Profile and optimize hot paths
2. **Multi-region support** — Cross-region cluster federation
3. **GPU scheduling v2** — Multi-GPU, fractional GPU allocation
4. **Web UI real-time** — WebSocket integration for live updates
5. **Operator pattern** — Kubernetes operator for cluster management

§5: Known Issues / Blockers

- None

§6: Quick Commands

Run all tests (requires GOWORK due to workspace modules)

```
GOWORK=$(pwd)/go.work go test ./... -race -count=1
```

Build all binaries

```
GOWORK=$(pwd)/go.work go build ./cmd/...
```

Build web UI

```
cd web && npm run build
```

Render Helm chart

```
helm template helix-cluster deploy/helm/
```

Run benchmarks

```
go test -bench=. ./test/benchmark
```

```
# Run chaos tests  
go test -race -count=1 ./test/chaos
```